

# Robust Adversarial Defenses in Federated Learning: Exploring the Impact of Data Heterogeneity

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**Abstract**—Federated Learning (FL) enables geographically distributed clients to collaboratively train machine learning models by exchanging local model parameters while preserving data privacy. In practice, FL faces two critical challenges. First, it is vulnerable to security issues as malicious clients would artificially harm the functionality of FL by launching poisoning attacks. Second, the inherent data heterogeneity among clients (termed Non-IID data in FL) naturally arises from distributed data ownership and significantly degrades model convergence and accuracy. However, with studies separately devoted to these two research lines, the interplay between data heterogeneity and security remains poorly understood. In this paper, we systematically investigate the relationship between data heterogeneity and adversarial robustness in FL. Specifically, we propose novel data partitioning algorithms that simulate Label-Conditional Non-IID and Feature-Conditional Non-IID with quantifiable heterogeneity levels. Further, we conduct extensive experiments to evaluate classical defense methods in the practical FL environment under state-of-the-art untargeted attacks. With results in various settings, we separately analyze the connection between Non-IID to defenses and attacks. Regarding attacks, with similar effects on models, Non-IID impacts the training in a different way compared with attacks. The interaction between attacks and Non-IID provides an opportunity to cause severe damage to FL. Regarding defenses, Non-IID induces heterogeneity in model distribution among clients which raises the difficulty of maintaining fidelity and robustness for defense methods.

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## I. INTRODUCTION

FEDERATED Learning (FL) is a decentralized machine learning paradigm that allows multiple clients to collaboratively train a shared global model without exposing their private data [1], [2], [3]. In FL, clients iteratively update local models using their proprietary datasets and transmit only model parameters—not raw data—to a central server for aggregation. This process preserves data privacy while reducing communication overhead, making FL particularly attractive for applications in healthcare, finance, and IoT systems.

The clients in FL share a common objective of training an effective machine learning model. However, mutual distrust prevents them from directly communicating their private data. These conflicting interests have led to the development of FL, but they have also exposed it to vulnerabilities. FL is susceptible to threats from adversaries who may disguise themselves as clients or hijack clients to disrupt the collaborative training process. These adversaries aim to manipulate local training to create poisoned models, which are then uploaded to the server, compromising the integrity of the global model. Such attacks launched by adversaries are named poisoning attacks.

Poisoning attacks in FL are typically categorized by their objectives: backdoor attacks, targeted attacks, and untargeted attacks. Backdoor attacks [4], [5], [6] aim to induce misclassification in the global model, either by introducing artificial backdoors or leveraging semantic backdoors. Targeted attacks [7], [8], [9] focus on causing the global model to misclassify a specific class of data. In contrast, untargeted attacks [10], [11], [12] seek to indiscriminately reduce the global model's accuracy across all tasks. This work focuses on untargeted attacks, as they represent a broader threat to FL systems by undermining the model's general utility—a critical requirement for real-world deployment [13].

Defending against poisoning attacks has become a cornerstone of FL research [4], [5], [6], [14]. Adversaries inject poisoned data into clients' local datasets or directly generate poisoned models to harm the functionality of FL. The former is known as data poisoning attacks [15] and the latter is known as model poisoning attacks [10], [11], [12]. To mitigate these threats, various defense methods have been proposed. Trimmed Mean [14] and Norm Bound [16] utilize comparison

and parameter limitation directly on model parameters to defend against poisoning attacks. CRFL [17] and FLMJR [18] delve deeper into the model's essence to enhance FL's robustness. However, these methods often assume IID data distributions among clients—an assumption rarely valid in practice.

While security issues are artificial challenges, data heterogeneity is a natural phenomenon that should be taken seriously in Federated Learning (FL) [2], [3]. In practice, the data distribution among clients varies, leading to Non-IID (Non-Independent and Identically Distributed) data. Theoretically results [3], [19] have demonstrated that the data heterogeneity impacts the convergence of the global model by increasing both local and global variances, thereby reducing the convergence rate. Empirical experiments have further shown that Non-IID data poses significant challenges to the learning accuracy of FL algorithms.

Existing research has primarily examined the impacts of data heterogeneity and poisoning attacks on FL in isolation. However, in practical scenarios, these two critical challenges often occur simultaneously, complicating the development of effective solutions. A comprehensive understanding of the interplay between data heterogeneity and security threats is essential for addressing these issues effectively. Therefore, this paper aims to explore the connection between data heterogeneity and security in FL.

In this study, we introduce a data partitioning algorithm designed to simulate quantifiable Non-IID scenarios in FL. We model each client's local data distribution as a conditional distribution of the underlying global data, focusing on two representative conditions: label-based and feature-based distributions. The Non-IID defined by these conditions are termed Label-Condition Non-IID and Feature-Condition Non-IID, respectively. To simulate these Non-IID conditions, we employ the Dirichlet distribution, adjusting its concentration parameter, denoted as  $\beta$ , to control the degree of heterogeneity. For label-based Non-IID, we apply the Dirichlet distribution directly to determine the distribution of labels across clients. For feature-based Non-IID, we extract feature characteristics shared by subsets of data to define the condition, using the Dirichlet distribution to determine the distribution of these feature characteristics among clients.

Further, we conducted extensive experiments simulating the cross-device environment to empirically investigate the relationship between security and data heterogeneity in FL. We examined this relationship from the perspectives of both attacks and defenses. Regarding attacks, we observed that both Non-IID data and untargeted attacks lead to accuracy degradation in the global model. We focused on comparing their impacts and exploring the effects when Non-IID data and attacks are simultaneously applied in FL. In terms of defenses, we examined the impact of Non-IID data on maintaining fidelity in scenarios without attacks and ensuring robustness in scenarios with attacks applied.

From our analysis of the empirical evaluation results, several key findings emerge. Considering attacks, while both Non-IID and untargeted attacks lead to similar accuracy degradation, Non-IID increases the difficulty of global model convergence.

This is in contrast to untargeted attacks, which aim to prevent convergence of the global model. The interaction between them can cause severe damage to the global model, although there is potential for even more harmful effects. As for defenses, Non-IID introduces heterogeneity into the model distribution among clients, making it challenging for defenders to distinguish malicious models. Consequently, Non-IID leads to degradation in the performance of defense methods, particularly those relying on comparisons among clients' local models, in terms of fidelity and robustness.

Our main contributions are as follows:

- We propose a novel data partitioning algorithm that facilitates the quantifiable simulation of Label-Condition Non-IID and Feature-Condition Non-IID in scenarios in FL. Specifically, we present an innovative approach to exploring the relationship between data in the embedding space, enabling the extraction of feature characteristics shared by subsets of data to define the feature condition.
- To evaluate the performance of defense and attack methods under Non-IID conditions, we conduct extensive experiments simulating practical production environments. We examine these scenarios from both attacker's and defender's perspectives, analyzing their interactions with data heterogeneity.
- Our findings reveal that the untargeted attacks evaluated exhibit distinct impacts on FL compared to Non-IID data, resulting in reduced stealthiness and ineffective interactions with Non-IID conditions. Additionally, we observe that Non-IID data introduces heterogeneity into the model distribution among clients, compromising the fidelity and robustness of defense methods.

## II. BACKGROUND

In this section, we introduce the background of this paper. Specifically, we present the system model of the Robust Federated Learning System, detailing each component. Following this, we review relevant literature on untargeted attacks and defense mechanisms, selectively highlighting the attack and defense methods employed in this study.

### A. System Model

In a typical FL system [1], distributed clients collaborate with a central server. Each client possesses a private dataset and performs local training to update its model. The server, often referred to as the parameter server, orchestrates the overall training process by aggregating the local models. In each round of training ( $t$ -th round), the server selects a subset of clients and sends the current global model  $w^t$  to them. The selected clients then perform local training to generate their local models  $w_i^t$ . The server collects these local models and aggregates them to produce the global model  $w^{t+1}$  for the next round of training ( $t + 1$ -th round).

In this paper, we focus on a specific scenario called Robust Federated Learning System (RFLS), in which the FL system faces security threats posed by adversaries. We assume that the adversaries can only launch attacks from the client side, while the server remains benign. Consequently, the RFLS consists

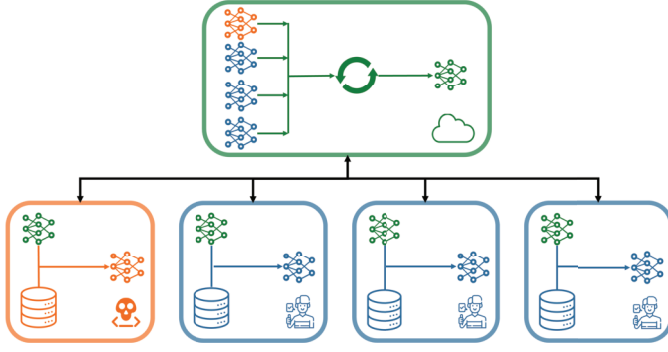


Fig. 1. The architecture of RFLS. The models in red, blue, and green represent the malicious model, the benign model, and the global model respectively. The squares in orange, blue, and green represent the malicious client, the benign client, and the server respectively.

of benign clients, adversarial clients, and a central server, as depicted in Figure 1. In the following sections, we will elaborate on the capabilities and objectives of each of these components.

1) *Server*: In addition to overseeing the training process of the FL system, the server in the RFLS—assumed to be benign—also plays a crucial role in defending against attacks launched by adversaries. A common defense strategy against poisoning attacks is to assess the quality of the uploaded models and identify malicious ones. However, model quality evaluation typically depends on estimating accuracy using a test dataset, which the server cannot directly access due to privacy constraints. Consequently, evaluating the performance of the uploaded models becomes a significant challenge. To address this, the server replaces the standard aggregation algorithm with a robust aggregation algorithm to mitigate the impact of attacks. In most cases, the server relies on the parameters of the uploaded models to infer their quality and distinguish between benign and malicious models.

2) *Adversaries*: Adversaries, also referred to as malicious clients, aim to launch poisoning attacks that compromise the utility of the global model. These attacks can be categorized based on their objectives: targeted attacks [7], [8], [9], which seek to make the model misclassify specific subsets or classes of data; backdoor attacks [4], [5], [6], which aim to induce misclassification in the model by exploiting artificially crafted or naturally occurring patterns; and untargeted attacks, which seek to maximize the model’s overall misclassification of data. This paper focuses on untargeted attacks [10], [11], [12], [20], as they pose a significant threat to all consumers of models trained via FL. Adversaries operate in a similar manner to benign clients, as each adversary has access to its own local dataset and can perform local training. Previous works have assumed that adversaries gain access to the local models of benign clients during training, allowing them to manipulate their malicious models. However, with the development of secure aggregation protocols, this assumption has become increasingly difficult to sustain in practice. Nevertheless, to fully evaluate the effectiveness of untargeted attacks, we retain this assumption in our empirical evaluation. The distributed nature of FL enables adversaries to collaborate, with the potential to hijack or masquerade as benign clients, thus

controlling approximately 10% to 20% of the total clients in the FL system.

### B. Attack Methods

A substantial body of research has focused on untargeted attacks in FL. These attacks aim to prevent model convergence and undermine the model’s utility across the underlying data distribution, posing significant challenges to FL systems. In this paper, we review four typical untargeted attacks including LIE [11], Fang [10], NDSS [12], and PoisonedFL [20]. We implement the first three attacks in our empirical evaluation.

1) *Lie*: LIE [11] is a well-known untargeted attack that aims to prevent the convergence of the global model. Adversaries establish a target aggregation direction in the parameter space that opposes the true convergence direction. By crafting updates that lie between the mean and the target direction, the malicious model appears closer to the mean than some benign models, which are positioned at the extremes of the distribution. This strategy enables attackers to bypass defense mechanisms and impede the convergence of the global model. Formally, the attacks can be expressed as:

$$z^{max} = \max_z \left( \phi(z) < \frac{n-m-s}{n-m} \right), s = \left\lfloor \frac{n}{2} + 1 \right\rfloor - m, \quad (1)$$

$$w_{i,j}^m = \text{mean}(w_{i,j}^b) - z^{max} \cdot \text{std}(w_{i,j}^b), \quad (2)$$

where  $n$  and  $m$  represent the number of total clients and malicious clients, respectively.  $\phi(z)$  is the Cumulative Standard Normal Function.  $w_{i,j}^b$  and  $w_{i,j}^m$  are the  $j$ -th parameter of the local updates of the  $i$ -th benign client and malicious client respectively. The mean and the standard deviation of benign clients can be captured by malicious clients leveraging some hijack tools. They can also be simulated by the malicious clients themselves.

2) *Fang*: Fang [10] is a model poisoning attack in which adversaries manipulate local updates to steer the global model in the opposite direction of convergence. Fang is specifically designed with targeted attacks tailored to different aggregation algorithms to maximize its stealthiness. Empirical results show that attacks optimized for specific aggregation algorithms can be transferred to others with minimal loss of effectiveness. Consequently, in this paper, we implement the Fang attack tailored for the Trimmed Mean aggregation method.

Specifically, considering the  $j$ -th dimension of the model parameters,  $s_j$  is set up to represent the changing direction of the global model.  $s_j = 1$  (or  $s_j = -1$ ) means that the  $j$ -th dimension of the parameter increases (decreases) upon the previous iteration.  $w_{max,j}$  and  $w_{min,j}$  denote the maximum and minimum of the  $j$ -th dimension of the local updates on benign clients, i.e.,  $w_{max,j} = \max \{w_{1,j}^b, w_{2,j}^b, \dots, w_{m,j}^b\}$  and  $w_{min,j} = \min \{w_{1,j}^b, w_{2,j}^b, \dots, w_{m,j}^b\}$ . The  $j$ -th dimension of the malicious model is formed as:

$$w_{i,j}^m \in [w_{max,j}, b \cdot w_{max,j}] (s_j = -1, w_{max,j} > 0), \quad (3)$$

$$w_{i,j}^m \in [w_{max,j}, w_{max,j}/b] (s_j = -1, w_{max,j} \leq 0), \quad (4)$$

$$w_{i,j}^m \in [w_{min,j}/b, w_{min,j}] (s_j = 1, w_{min,j} > 0), \quad (5)$$

$$w_{i,j}^m \in [b \cdot w_{min,j}, w_{min,j}] (s_j = 1, w_{min,j} \leq 0), \quad (6)$$

where the  $j$ -th dimension of the malicious model  $w_{i,j}^m$  is randomly sampled in a fixed range determined by  $w_{max,j}$ ,  $w_{min,j}$  and  $s_j$ .

3) **NDSS**: NDSS [12] is a comprehensive framework for model poisoning attacks in federated learning (FL). It frames the objective of untargeted attacks as an optimization problem and generates malicious models by solving this problem. NDSS is equipped with multiple attack strategies to bypass various defense mechanisms across different scenarios. In this paper, we select the MinMax attack as a representative example.

The construction of the NDSS attack can be likened to solving an optimization problem.:

$$\operatorname{argmax}_{\gamma} \max_i \|w^m - w_i^b\|_2 \leq \max_{i,j} \|w_i^b - w_j^b\|_2, \quad (7)$$

$$w^m = w^b - \gamma \cdot w^p, w^p = -\frac{w^b}{\|w^b\|_2}, w^b = f_{avg}(w_i^b), \quad (8)$$

Accordingly,  $f_{avg}$  denotes the standard aggregation algorithm FedAvg. The attack aims to generate a malicious update that is close to benign updates, and meanwhile, points in the inverse direction to the benign updates.

4) **PoisonedFL**: PoisonedFL [20] is a novel model poisoning attack against federated learning that leverages multi-round consistency to maximize attack effectiveness without requiring knowledge of genuine clients' data or defenses. Unlike existing attacks, which suffer from self-cancellation due to inconsistent malicious model updates across rounds, PoisonedFL crafts malicious updates with a consistent random sign vector and dynamically adjusted magnitude vectors. This ensures the global model is progressively moved along a random direction, leading to high testing error rates. The attack adapts to unknown defenses by estimating genuine client updates from past global model differences and using statistical hypothesis testing to adjust the magnitude.

### C. Defense Methods

The literature has explored various approaches to enhancing the robustness of FL against poisoning attacks. We categorize the defense methods against poisoning attacks and select a representative method from each category for our empirical evaluations.

1) **Vector-Wise Comparison**: An intuitive approach to defending against untargeted attacks is to distinguish and eliminate malicious clients from benign ones. From the perspective of the server-side defender, comparison-based methods provide a natural way to achieve this goal. Since the server does not have direct access to the clients' data, treating local updates as vectors and performing vector-to-vector comparisons is an effective way to identify malicious updates. Consequently, early FL security research proposed a series of defense methods based on vector-wise comparisons [14], [21], [22], [23], [24]. In this paper, we review Trimmed Mean [14], FedRoLa [25] and FedREDefense [26]. We implement Trimmed Mean for further empirical evaluations.

**Trimmed Mean** Trimmed Mean [14] is an adaptation of the traditional Byzantine algorithm for Federated Learning (FL).

In this approach, for each dimension of the global model, the server receives local updates from clients, excludes the largest and smallest  $k$  values for that dimension, and calculates the average of the remaining values to determine the final result for that dimension. Although Trimmed Mean sacrifices some model utility, empirical results demonstrate its effectiveness in defending against basic poisoning attacks.

**FedRoLa** FedRoLa [25] is a novel defense mechanism designed to protect FL from model poisoning attacks without relying on historical or validation data. It employs a layer-based approach, selecting critical neural network layers for analysis and using similarity metrics—Layer Alignment Similarity Index and Peer Consensus Similarity Index to detect malicious client updates. A voting mechanism across layers minimizes false positives, while an adaptive discount factor balances security and performance by weighting potentially malicious updates.

**FedREDefense** FedREDefense [26] is a novel defense mechanism designed to protect FL from model poisoning attacks by assessing the authenticity of client model updates without relying on cross-client or global information, making it robust to NonIID and high numbers of malicious clients. It leverages the insight that genuine model updates from benign clients can be approximately reconstructed using distilled local knowledge, while malicious updates from hand-crafted poisoning attacks cannot. In each training round, FedREDefense employs a bi-level optimization process to distill compact local knowledge (parameterized as 10 data-label pairs) for each client, reconstructs the client's model update, and calculates the reconstruction error. Updates with errors exceeding a predefined threshold are flagged as malicious and excluded from aggregation, with malicious clients removed from future rounds.

2) **Vector-Wise Bounding**: Another approach to defending against untargeted attacks is to restrict and tolerate the uploaded models from malicious clients. This strategy is implemented through Vector-wise Bounding, where the defender establishes specific conditions for the clients' local models. Only models that satisfy these conditions are permitted to participate in the aggregation process. These conditions can either be predetermined before training or dynamically assessed during the training process. Vector-wise Bounding has been extensively studied in academic research and applied in industrial settings [13], [16], [27]. In this paper, we select Norm Bound as a representative example.

**Norm Bound** Namely, Norm Bound aims to limit the behavior of the clients by bounding the vector norm of the local updates. References [13], [16], and [27] Specifically, Norm Bound can be treated as a variation of FedAvg since it induces a norm clipping before calculating the average of clients' local updates. It forms as:

$$w = \frac{1}{n} \sum_{i=1}^n \operatorname{Clip}(w_i), \quad (9)$$

$$\operatorname{Clip}(w_i) = w_i \cdot \min\left(1, \frac{\|w_i\|_2}{p}\right), \quad (10)$$

Accordingly, the norm of local updates is bounded by  $p$ . In practice,  $p$  can be statically determined before learning or dynamically estimated during the training. The norm is typically estimated using the L2 norm. Norm Bound is widely utilized in practice due to its efficiency in defending against model poisoning attacks. Its implementation is straightforward, and it can be combined with other cryptographic tools to further enhance the privacy preservation of FL.

3) *Theoretical Certification*: While defense methods have shown empirical effectiveness, theoretical certification of robustness is equally crucial, particularly given the unpredictable nature of attackers. Recent research has increasingly leveraged the functionality of local models to provide theoretical guarantees for security in FL [17], [28]. In this paper, we select CRFL as a representative example for our analysis.

**CRFL** CRFL [17] is an FL framework designed to ensure robustness during both the training and inference phases. In CRFL, the server-side framework computes the average of local updates, similar to FedAvg. Subsequently, the global model parameters are clipped to ensure their norm remains bounded. Additionally, CRFL introduces isotropic Gaussian noise directly into the aggregated global model parameters. Formally, the norm clipping and noise injection can be expressed as:

$$\text{Clip}(w) = w \cdot \min\left(1, \frac{\|w\|_2}{\rho}\right), \quad (11)$$

$$\text{Perturb}(w) = w + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2 \mathbb{I}), \quad (12)$$

where  $w$  is the global model. *Clip* bound the norm of global model  $w$  by  $\rho$ . The noise added by *Perturb* is sampled on Gaussian distribution  $\mathcal{N}(0, \sigma^2 \mathbb{I})$ .  $\rho$  and  $\sigma$  can be dynamically tuned during the training. During the inference phase, the server applies randomized parameter smoothing to the final model and generates predictions based on the smoothed parameters. Theoretically, CRFL ensures that the trained global model is certifiably robust against backdoor attacks, provided that the backdoor remains within certain certified bounds.

**FoundationFL** FoundationFL [28] is a defense mechanism to enhance the robustness of FL against poisoning attacks without designing new Byzantine-robust aggregation rules. Instead, it strengthens existing rules like Trimmed-mean and Median. In FoundationFL, the server generates synthetic model updates in each training round based on clients' local model updates. It calculates a closeness score for each client's update to identify the one least aligned with extreme values, using this as the basis for synthetic updates. These synthetic updates are combined with clients' updates and aggregated using Trimmed mean or Median rules. This approach reduces variance in updates, mitigating the impact of malicious outliers. The method is theoretically proven to converge under Byzantine settings and is validated through extensive experiments.

4) *Client-Side Defense*: Defense mechanisms can be implemented not only on the server side but also on the client side. Unlike the server, clients have access to their own data. Benign clients can serve as defenders [18], [29], utilizing their private data to generate reliable results for local update

comparison and bounding. In this paper, we select FLMJR as a representative example.

**FLMJR** FLMJR [18] is an FL method that focuses on enhancing robustness through local training on the client side. Specifically, FLMJR modifies the local objective function of the clients. In addition to the native training objective, a model-output Jacobian regularization term is incorporated into the clients' local objectives. It forms as:

$$\mathcal{L}_i(w_i) = \mathcal{L}_{\text{local}}(w_i) + \lambda \mathcal{L}_{\text{MJR}}(w_i), \quad (13)$$

$$\mathcal{L}_{\text{MJR}}(w_i) = \sum_{j,c} \left[ \frac{\partial f_c(w_i)}{\partial w_{i,j}} \right]^2, \quad (14)$$

Accordingly,  $\mathcal{L}_i$ ,  $\mathcal{L}_{\text{local}}$  and  $\mathcal{L}_{\text{MJR}}$  represent the local objective, native training objective, and Jacobian regularization respectively.  $f_c(w_i)$  is the  $c$ -th dimension of the output of the local model  $w_i$ .  $w_{i,j}$  is the  $j$ -th dimension of the parameter of local model. With the support of benign clients, FLMJR improves the model's stability against perturbations in the parameter space, thereby enhancing robustness. As a defense method applied on the client side, FLMJR offers the flexibility to be combined with the general aggregation method FedAvg as well as other robust aggregation techniques.

### III. SIMULATING DATA HETEROGENEITY IN FEDERATED LEARNING

In the early stages of FL development, it was often assumed that clients' data distributions were homomorphic, meaning that the data from clients followed an independent and identically distributed (IID) pattern. However, this assumption does not hold in practice. In real-world scenarios, clients' data distributions are heterogeneous, resulting in non-independent and non-identically distributed (Non-IID) data. This variation in data distribution among clients impedes the effectiveness of FL, leading to slower convergence and reduced generalization of the model. In this section, we define Non-IID as the conditional distribution of the underlying data and identify two representative forms of Non-IID. Additionally, we introduce a data partition algorithm that simulates quantifiable Non-IID.

#### A. Quantifying Non-IID

Considering classification tasks, a single data sample comprises a feature  $x$  and its corresponding label  $y$ . The data is drawn from an underlying data distribution denoted as  $P(x, y)$ . In FL, there is an implicit assumption that all local data owned by clients are drawn from this underlying data distribution that enables the generalization among data and guarantees the basic functionality of FL. The local data distribution of the  $i$ -th client is denoted as  $P_i(x, y)$ . Ideally, all local data distributions of clients would be uniformly drawn from the underlying data distribution, i.e.  $P_i(x, y) = P(x, y), i = 1, \dots, n$ , resulting in a homomorphic distribution among clients. However, in practice, the data distribution among clients is heterogeneous as each client's local data distribution diverges from others'.

In such scenarios, the local data distribution can be viewed as a conditional distribution of the underlying data, where the condition is determined by various attributes, such as the

size or quality of the dataset. In this paper, we focus on two representative scenarios: Label-Condition Non-IID and Feature-Condition Non-IID, which are defined by label-based and feature-based conditions, respectively. Formally, in Label-Condition Non-IID  $P_i(x, y) = P(x | y)P_i(y)$ ,  $i = 1, \dots, n$ , while in Feature-Condition Non-IID  $P_i(x, y) = P(y | x)P_i(x)$ ,  $i = 1, \dots, n$ .

Previous studies have predominantly focused on Label-Condition Non-IID. Consistent with these approaches, we use a characterized Dirichlet distribution to simulate Label-Condition Non-IID. However, Feature-Condition Non-IID has received less attention, primarily due to the difficulty in defining feature conditions on general datasets. To address this gap, we have developed a novel algorithm that extracts feature characteristics shared by subsets of data, enabling the simulation of Feature-Condition Non-IID.

### B. Simulating Label-Condition Non-IID

To simulate Non-IID on a general dataset in FL, we divide the dataset into subsets and assign them to clients. Alternatively, one could use real FL datasets, as they are already partitioned into heterogeneous subsets. However, the heterogeneity of real FL datasets is challenging to quantify and characterize. Therefore, in this paper, we choose to split the general dataset into subsets to simulate Non-IID.

In Label-Condition Non-IID, the label conditions  $P_i(y)$  vary across clients. For example, in a hand-written digit recognition task, some clients may predominantly write “1” and “2”, while others may focus on “3” and “4”. To simulate quantifiable Label-Condition Non-IID, we adopt a characterized Dirichlet distribution, following previous studies. The Dirichlet distribution offers several advantages for this purpose. First, it provides flexibility in generating label distributions, allowing a wide range of label conditions to be simulated. Second, it scales efficiently with a large number of labels and clients, making it well-suited for simulating Non-IID in FL scenarios with diverse label distributions across clients. Moreover, the statistical properties of the Dirichlet distribution align with the characteristics of label distributions in Non-IID settings, making it an appropriate choice for simulation. Finally, the parameters of the Dirichlet distribution are interpretable, offering insights into the simulated label conditions and their impact on FL.

The data partition algorithm based on Dirichlet distribution is outlined in Algorithm 1. For each label, we sample a proportion  $Prop$  from the Dirichlet distribution  $Dirichlet(\beta)$  to allocate data with the corresponding label to clients. Here,  $\beta$  represents the concentration parameter of Dirichlet distribution. A smaller value of  $\beta$  induces more heterogeneity among clients’ data distributions. In addition to using the Dirichlet distribution, we implemented a balancing mechanism to prevent the over-allocation of data to certain clients. To ensure that each client has a sufficient amount of data, we also set a minimum data size threshold. The partition algorithm generates an index map that associates clients with their respective data subsets. Using this index map, we then distribute the data to clients, thereby constructing the FL system.

### Algorithm 1 Data Partition Algorithm Label-Condition Non-IID Feature-Condition Non-IID

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**Input:**  $D$ - the general dataset;  $N$ - number of clients;  $\beta$ - concentrate parameter of Dirichlet Distribution;  $Size_{min}$ -minimum size of the clients local dataset.

**Output:**  $I$ -index map for clients

- 1 Set  $K$  as the number of Labels.
- 2 Set  $K$  as the number of Feature characters.
- 3  $S_{min}=0$ .
- 4 **while**  $S_{min} < Size_{min}$  **do**
- 5   Initialize index map:
- 6   **for**  $j = 1, 2, \dots, n$  **do**
- 7      $I_j = \{\}$
- 8   **end**
- 9   **for**  $k = 0, 1, \dots, K - 1$  **do**
- 10     Collect the data with label equals to  $k$ :
- 11      $Ind_k = \{i \mid y_i = k\}$
- 12     Collect the data with feature characteristic equals to  $k$ :  $Ind_k = \{i \mid z_i = k\}$
- 13     Sample proportion from Dirichlet distribution:
- 14      $Prop \sim Dirichlet(\beta)$
- 15     Balance the proportion:
- 16     **for**  $j = 1, 2, \dots, n$  **do**
- 17       **if**  $\|I_j\| > \|D\|/n$  **then**
- 18          $Prop_j = 0$
- 19       **end**
- 20     **end**
- 21     Normalize the proportion:
- 22      $Prop = Prop / \sum_j Prop_j$
- 23     Distribute data:  $temp = 0$
- 24     **for**  $j = 1, 2, \dots, n$  **do**
- 25       **for**  $i = temp, \dots, temp + \|Ind_k\| * Prop_j$  **do**
- 26         Append  $Ind_k[i]$  into  $I_j$
- 27       **end**
- 28        $temp = temp + \|Ind_k\| * Prop_j + 1$
- 29     **end**
- 30      $S_{min} = \min_j \|I_j\|$
- 31 **end**

---

### C. Simulating Feature-Condition Non-IID

Similar to Label-Condition Non-IID, in Feature-Condition Non-IID, the feature conditions  $P_i(x)$  differ across clients. For instance, in the MNIST classification task, clients may write digits in different styles, such as using thicker or thinner strokes. However, simulating Feature-Condition Non-IID presents significant challenges. In Label-Condition Non-IID, the label is a scalar, making it straightforward to define the conditions. In contrast, in Feature-Condition Non-IID, the feature is a high-dimensional vector, which complicates the direct definition of conditions based on the feature. To overcome this challenge, we propose a novel algorithm that extracts feature characteristics shared by subsets of data and utilizes these characteristics to define feature conditions.

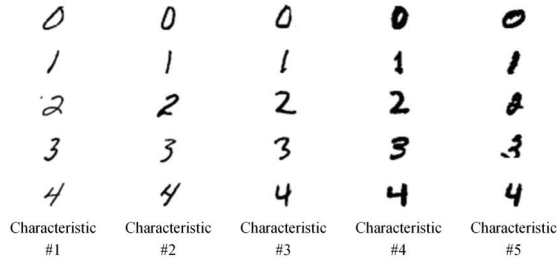


Fig. 2. An example of feature characteristics in MNIST dataset. Each column contains 5 figures which share the same feature characteristics.

We describe the process of extracting feature characteristics in detail. First, we feed all the data into a pre-trained model to obtain their embeddings. Next, based on the labels of the data, we divide the embeddings into  $K$  subsets, ensuring that each subset contains data points with the same label. For each subset, we apply a Gaussian Mixture Model to partition the embeddings into clusters. To determine the optimal number of clusters, we estimate the average silhouette value, which measures the quality of the clustering. We select the largest number of clusters with a silhouette value greater than 0, ensuring the validity of the clusters and providing diversity in the feature characteristics.

Subsequently, we perform multi-party matching to group clusters across labels and generate the feature characteristics. We compute an optimal multi-party cost matrix, which is defined by the Kullback-Leibler (KL) divergence between clusters. We then progressively perform bi-party matching for randomly selected pairs to simulate multi-party matching. Finally, the clusters are grouped into new sets, which we define as feature characteristics.

The fundamental rationale behind our feature characteristics extraction is that, in the embedding space, data points that are geographically close tend to form clusters with similar feature characteristics. We present an example of the extracted feature characteristics in Figure 2 to demonstrate the effectiveness of our method. Furthermore, we leverage the Dirichlet distribution to simulate Feature-Condition Non-IID, with feature conditions defined by these feature characteristics.

#### IV. SETUP OF EMPIRICAL EVALUATION

In this section, we present the setup of our empirical evaluation, including the dataset and the configuration of the FL system.

##### A. Dataset

1) *MNIST*: The MNIST dataset [30] is a widely recognized benchmark for handwritten digit recognition. It contains 60,000 training images and 10,000 test images, each a grayscale image of size  $28 \times 28$  pixels. This dataset is commonly used to train machine learning models, especially in the field of computer vision, to recognize handwritten digits from 0 to 9.

2) *CIFAR10*: The CIFAR-10 dataset [31] is another widely used benchmark for image classification tasks. It consists of 60,000 color images of size  $32 \times 32$  pixels, divided into

10 classes, with 6,000 images per class. The classes include airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. This dataset is commonly employed to evaluate machine learning algorithms, particularly in deep learning, for image classification tasks.

##### B. Federated Learning Setting

To simulate the cross-device scenario, in our empirical evaluation, we construct the FL system with 500 clients in our empirical evaluation. In each communication round, 10% of the clients are randomly selected to participate in the training. For local training, the selected clients perform 5 step of stochastic gradient descent (SGD) to update their local models. The updated local models are then uploaded to the server for aggregation. The learning rate of the local training is set to 0.1. For CIFAR10 classification, we use ResNet8 [32] as the network architecture and train for 2000 communication rounds. For MNIST classification, we use LeNet [30] as the network architecture and train for 1000 communication rounds. By default, we set the malicious clients' ratio to 10%. Our code is implemented through the deep learning framework Pytorch. We utilize 2 NVIDIA V100 graphics cards to run the experiments.

#### V. CONNECTION BETWEEN ATTACK AND DATA HETEROGENEITY

While untargeted attacks are intentional, aiming to disrupt the model through malicious actions by adversaries, data heterogeneity is an inherent challenge that FL must contend with. Given their similar effects on FL, this section empirically investigates the relationship between Non-IID and untargeted attacks. Specifically, we aim to address the following questions:

- *What are the distinguishing features between Non-IID and untargeted attacks, despite their similar effects of accuracy decline?*
- *How do untargeted attacks interact with Non-IID? Can adversaries exploit this interaction to launch more potent attacks and inflict greater damage on FL?*

To address these research questions, we conducted a series of comprehensive experiments and meticulously analyzed the results in the subsequent subsections. Our experimental framework is structured into two distinct components: a comparative analysis of attacks and Non-IID scenarios, and an in-depth exploration of the interplay between attacks and Non-IID conditions.

##### A. Comparison Between Attacks and Non-IID

We begin by examining the effects of Non-IID and untargeted attacks when applied separately to FL. Both factors contribute to a decline in model accuracy, but our goal is to understand the differences in their impacts on FL. Training loss and test accuracy are common metrics for evaluating model performance, reflecting its convergence and generalization capabilities. In this section, we focus on training loss to assess the impact of Non-IID and untargeted attacks. Training

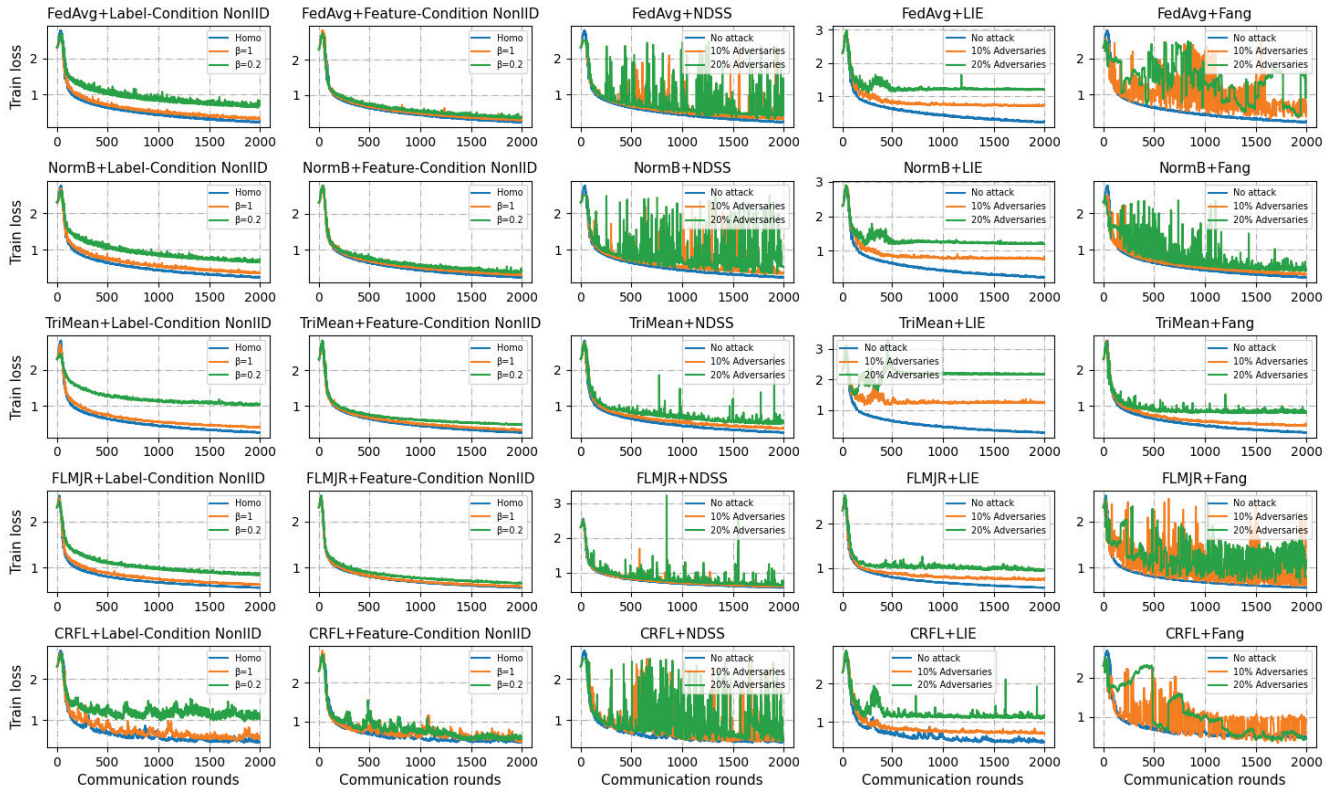


Fig. 3. The experiment results of comparison between untargeted attacks and Non-IID. From top to bottom, the figures in the first to fifth rows represent the results on FedAvg, Norm Bound, Trimmed Mean, FLMJR, and CRFL, respectively. From left to right, the figures in the first to fifth columns represent the results on Label-Condition Non-IID, Feature-Condition Non-IID, NDSS, LIE and Fang respectively.

loss is a suitable metric for several reasons. First, untargeted attacks often aim to prevent convergence, making training loss a more relevant metric than test accuracy for evaluating the attackers' objectives. Second, as a defender, it is challenging to collect sufficient data to estimate test accuracy, while training loss can be easily obtained by collecting local training loss values from clients based on their respective datasets.

Specifically, we conduct experiments on the CIFAR10 dataset. As we mentioned earlier, we apply four defense methods including FedAvg, Norm Bound, Trimean, and CRFL, alongside the standard aggregation algorithm, FedAvg. LIE, Fang, and NDSS are implemented to represent untargeted attacks. First, we perform FL in the IID scenario without any attacks, treating this as the baseline for the experiment. Next, we conduct FL in both Label-Condition Non-IID and Feature-Condition Non-IID scenarios, without any attacks, to examine the impact of Non-IID. We tune  $\beta$  to 1 and 0.2 to simulate varying degrees of data heterogeneity. Finally, we perform FL in IID scenarios with three different attacks to evaluate the impact of untargeted attacks. The ratio of adversaries is set to 10% and 20% to observe the effect of attacks in different settings. The experimental results are shown in Figure 3.

We first focus on the results in the first two columns (from left to right) of the figures and analyze the impact of Non-IID on FL. In both IID and Non-IID scenarios, the model consistently converges after sufficient rounds of training, indicating that the convergence process remains intact. However,

the final convergence point differs between the scenarios, with Non-IID resulting in a higher training loss compared to the IID scenario. The introduction of Non-IID data increases heterogeneity, leading to larger training losses. This effect is particularly pronounced in the Label-Condition Non-IID scenario, especially when using the Trimmed Mean as the aggregation algorithm. Similar trends are also observed in the Feature-Condition Non-IID scenario.

**Tip 1: Label-condition Non-IID and Feature-condition Non-IID do not prevent the convergence of the global model but lead the model to converge to the status with a higher train loss.**

Furthermore, we focus on the performance of LIE attack. Accordingly, when the LIE attack is applied, the training loss curves diverge from those seen with Fang and NDSS. Initially, the training loss surges during the early stages of training. However, after several rounds, the training loss remains persistently high and shows no signs of decreasing further. This behavior underscores the vulnerability of LIE in FL, as it prevents the model from converging to an optimal state. Consequently, LIE attack achieves a similar impact as Non-IID as they have a similar influence on the convergence of FL. However, The utility of Non-IID is stealthy as it only affects the convergence point of FL whereas LIE would stop the convergence at a certain point.

**Tip 2: LIE attack are reflected to prevent the convergence as it results in a relatively stable convergence curve,**

**causing the model to maintain a state with consistently higher training loss.**

Then, we consider the impact of NDSS attack and Fang attack. When applying NDSS and Fang, we observe fluctuations in the model's convergence curves. Throughout the training process, the training loss experiences sudden and significant increases and decreases. The increases are caused by Fang and NDSS, while the subsequent decreases are attributed to regularization effects from the local training of benign clients. In most cases, by the end of the training, the training loss reaches a low value similar to scenarios where no attack is applied, following sufficient rounds of training. These two attacks are significantly different from Non-IID. Reflected by the results, these attacks attempt to prevent convergence aggressively in each round of training and can be easily reversed by the benign clients. As a result, the loss curves are unstable and the convergence endpoint would not significantly change.

**Tip 3: NDSS and Fang are also reflected to prevent the convergence of the global model. They cause a sudden increase in training loss during the training process but do not ultimately prevent convergence.**

Building upon the experiment results, it is obvious that Non-IID has a different impact on FL compared to untargeted attacks. While both lead to a similar decline in model accuracy, Non-IID and untargeted attacks affect training loss in distinct ways. Label-Condition Non-IID and Feature-Condition Non-IID primarily shift the model's convergence destination. In contrast, NDSS and Fang primarily lead to increases in training loss, while LIE introduces fluctuations in the loss and prevents it from converging to a low value.

Considering the reason for the difference, recent theoretical research [3], [19], [33] suggests that the impact of Non-IID can be viewed as an increase in both local and global variances in FL. This increase in variance poses challenges and raises the difficulty of convergence. This is reflected in the training loss, which, although decent, shows a slower decrease. Regarding untargeted attacks, NDSS and Fang are designed to steer the global model towards the opposite direction of convergence, leading to sudden and significant increases in train loss during training. In contrast, LIE targets the model distribution of clients by manipulating local models, creating a deadlock with benign clients and preventing the model from converging.

As previous studies have shown, untargeted attacks often go undetected for extended periods due to the minimal reduction in accuracy, which is difficult to identify in practice. Moreover, even if the accuracy decline is detected, it is challenging for the server to determine its cause, as Non-IID can also lead to similar accuracy reductions. However, our evaluation demonstrates that untargeted attacks affect train loss differently than Non-IID, which could serve as a key indicator for their detection. Train loss can be estimated by benign clients based on their local data, and the server can access this information through trusted clients. Our results show that by monitoring train loss, untargeted attacks can be distinguished from Non-IID, as they produce distinctly different impacts.

## B. Interaction Between Attacks and Non-IID

In this section, we examine the scenario where both Non-IID and untargeted attacks are simultaneously applied to the FL system. From the perspective of the adversaries, we aim to investigate whether the attacks can exploit Non-IID to induce a greater decline in model accuracy. To clearly illustrate the impact of both Non-IID and untargeted attacks, we estimate the model's test accuracy in this section. Specifically, we focus on the accuracy decline caused by untargeted attacks, comparing it to the scenario in which no attacks are applied.

Specifically, we conduct experiments using the CIFAR10 dataset. Due to space limitations, we select Trimmed Mean, Norm Bound, CRFL, and FedAvg as the aggregation algorithms. LIE, Fang, and NDSS are instantiated as untargeted attacks, with 10% of the clients set as adversaries. For each attack-aggregation algorithm pair, we simulate five data distribution scenarios: homomorphic distribution, Label-condition Non-IID with  $\beta = 1$ , Label-condition Non-IID with  $\beta = 0.2$ , Label-condition Non-IID with  $\beta = 1$ , Feature-condition Non-IID with  $\beta = 1$  and Feature-condition Non-IID with  $\beta = 0.2$ . For each scenario, we estimate the test accuracy per round both with and without adversaries. The results of the experiments are presented in Figure 4.

First, we focus on the second column of the figures, which displays the experimental results with LIE applied. LIE is effective in causing accuracy declines in FL. The results show that, across all evaluated defense methods, LIE leads to a significant decrease in accuracy in IID scenarios. Compared to IID scenarios, LIE proves to be more effective in Non-IID scenarios, as it induces a greater accuracy decline in the model. Specifically, in Label-Condition Non-IID and Feature-Condition Non-IID scenarios, the attack's effect is more pronounced compared to IID scenarios, particularly when Trimmed Mean is used as the defense method. As the degree of Non-IID increases, the impact of the attack also intensifies.

**Tip 4: LIE causes obvious damage to FL with all the evaluated defense methods. Furthermore, the interaction with Label-Condition Non-IID and Feature Condition Non-IID benefits LIE to cause more harmful damage.**

We now discuss the impacts of Fang and NDSS. The results show that both NDSS and Fang have minimal effects. Fang notably reduces accuracy in FedAvg and CRFL in IID scenarios, but its impact on Trimmed Mean and Norm Bound is negligible. NDSS performs even worse, showing unsatisfactory results across all defense methods. When both Non-IID and untargeted attacks are applied simultaneously in FL, Non-IID generally amplifies the effects of these attacks. Specifically, the accuracy declines for both attacks are more pronounced in scenarios with Label-Condition Non-IID and Feature-Condition Non-IID. However, Non-IID does not enhance the performance of NDSS when CRFL is used as the defense method. Similarly, Non-IID does not benefit Fang or NDSS when Trimmed Mean is used as the defense method.

**Tip 5: Fang and NDSS do not provide satisfactory performance in attacking FL. The interaction with two types of Non-IID raises their effects of attacks. However, the benefit from Non-IID is insufficient to damage the functionality of the model.**

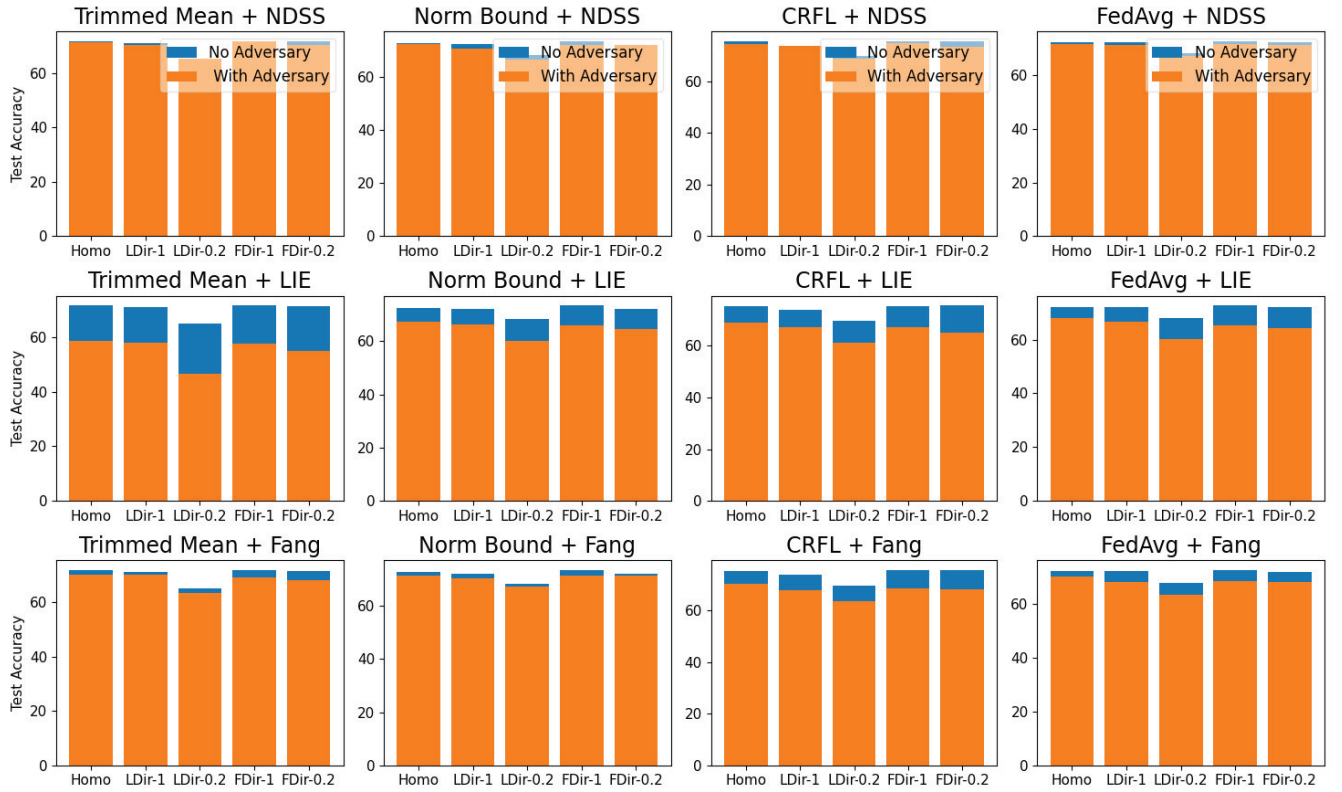


Fig. 4. The experiment results of interaction between untargeted attacks and Non-IID. From top to bottom, the figures in the first to third rows represent the results on NDSS, LIE, and Fang respectively. From left to right, the figures in the first to fourth columns represent the results on Trimmed Mean, Norm Bound, CRFL, and FedAvg respectively. LDir and FDir represent Label-Condition Non-IID and Feature-Condition Non-IID respectively. The following number is the value of  $\beta$ .

According to our experimental results, the interaction between Non-IID and untargeted attacks benefits adversaries, as the effects of untargeted attacks are consistently amplified by Non-IID. In the case of LIE, which performs notably well in IID scenarios, this interaction with Non-IID further boosts its effectiveness, resulting in significant damage to FL. For NDSS and Fang, although they generally perform poorly in all scenarios, their interaction with Non-IID still enhances their effectiveness in most cases.

The interaction with Non-IID significantly benefits LIE, while Fang and NDSS experience only slight advantages. This discrepancy can be attributed to LIE's reliance on data heterogeneity. Non-IID causes divergence among clients' local models, which LIE can exploit to further disrupt the model distribution. As a result, the aggregated global model struggles to converge. In contrast, Fang and NDSS are designed to steer the global model away from convergence, a strategy less affected by Non-IID. Since Non-IID increases the difficulty of convergence, Fang and NDSS do not benefit from it as much as LIE does.

## VI. CONNECTION BETWEEN DEFENSE AND DATA HETEROGENEITY

In this section, we examine the defender's role in FL and its relationship with data heterogeneity. The defender's responsibilities primarily encompass two key aspects: (1) Ensuring the convergence of the global model in the absence of adversaries, which we refer to as fidelity. This

involves maintaining the model's accuracy and reliability. (2) Ensuring that the global model remains resistant to attacks in the presence of adversaries, which we refer to as robustness. This entails safeguarding the model from malicious manipulations while preserving its performance under adversarial conditions. Given that Non-IID contributes to the performance degradation of FL, this section aims to empirically explore the connection between Non-IID and defenses. Specifically, we seek to address the following questions:

- How does Non-IID affect the fidelity of defense methods in the absence of untargeted attacks?
- How does Non-IID influence the robustness of defense methods when untargeted attacks are applied?

To address these questions, we conduct comprehensive evaluations of state-of-the-art defense methods. Our experiments are divided into two parts: evaluating the fidelity of defense methods in Non-IID scenarios and assessing their robustness in the presence of Non-IID.

### A. Impacts of Non-IID to the Fidelity of Defenses

Firstly, we evaluate the performance of defense methods in Non-IID scenarios without the application of any attacks. Defenders must strike a balance between robustness against poisoning attacks and the preservation of utility, ensuring the model retains a high level of accuracy. Non-IID data distribution often results in a decline in the accuracy of the global FL model, presenting a challenge for defenders in maintaining

TABLE I

TEST ACCURACY OF DEFENSE METHODS IN IID, LABEL-CONDITION NON-IID, AND FEATURE-CONDITION NON-IID WITHOUT ATTACKS APPLIED

| CIFAR10      |       |                         |             |             |                           |             |             |
|--------------|-------|-------------------------|-------------|-------------|---------------------------|-------------|-------------|
| Method       | IID   | Label-Condition Non-IID |             |             | Feature-Condition Non-IID |             |             |
|              |       | $\beta=1$               | $\beta=0.6$ | $\beta=0.4$ | $\beta=1$                 | $\beta=0.6$ | $\beta=0.4$ |
| FedAvg       | 72.24 | 72.21                   | 71.9        | 70.95       | 72.73                     | 72.49       | 72.69       |
| Trimmed Mean | 71.76 | 71.09                   | 70.44       | 69.42       | 71.75                     | 71.78       | 70.89       |
| NormBound    | 72.59 | 72.08                   | 71.73       | 71.0        | 73.43                     | 72.71       | 72.55       |
| CRFL         | 75.34 | 73.86                   | 73.50       | 72.66       | 75.37                     | 74.92       | 74.79       |
| FLMJR        | 76.59 | 75.85                   | 74.94       | 74.13       | 75.10                     | 75.24       | 74.26       |

| MNIST        |       |                         |             |             |                           |             |             |
|--------------|-------|-------------------------|-------------|-------------|---------------------------|-------------|-------------|
| Method       | IID   | Label-Condition Non-IID |             |             | Feature-Condition Non-IID |             |             |
|              |       | $\beta=1$               | $\beta=0.6$ | $\beta=0.4$ | $\beta=1$                 | $\beta=0.6$ | $\beta=0.4$ |
| FedAvg       | 98.84 | 98.79                   | 98.81       | 98.82       | 98.17                     | 97.97       | 98.13       |
| Trimmed Mean | 98.58 | 98.34                   | 98.19       | 98.17       | 97.13                     | 96.97       | 96.91       |
| NormBound    | 98.90 | 98.86                   | 98.88       | 98.78       | 97.96                     | 97.91       | 97.96       |
| CRFL         | 98.92 | 98.75                   | 98.84       | 98.83       | 98.03                     | 98.17       | 98.05       |
| FLMJR        | 99.06 | 99.06                   | 98.92       | 98.92       | 98.6                      | 98.63       | 98.61       |

fidelity. The goal of this subsection is to explore the fidelity of defense methods in Non-IID scenarios by estimating the test accuracy of the global model across various situations, thereby directly illustrating the fidelity of the defense methods.

Specifically, we conducted experiments on the CIFAR10 dataset and MNIST dataset. We implement 4 state-of-the-art defense methods: Trimmed Mean, Norm Bound, CRFL, FLMJR along with the standard aggregation method FedAvg. We evaluate the performance of each method in seven scenarios: IID, Label-Condition Non-IID with  $\beta = \{1, 0.6, 0.4\}$  and Feature-Condition with  $\beta = \{1, 0.6, 0.4\}$ . No attack is applied in these experiments as we aim to evaluate the performance of the defense methods in purely Non-IID scenarios. The results are shown in Table I.

Firstly, we focus on the second column of the figures, which displays the experimental results with LIE applied. LIE proves effective in causing declines in accuracy within FL. The results show that, across all evaluated defense methods, LIE leads to a significant drop in accuracy in IID scenarios. In comparison to IID scenarios, LIE proves even more effective in Non-IID scenarios, causing a greater decline in accuracy. Specifically, in scenarios involving Label-Condition Non-IID and Feature-Condition Non-IID, the impact of the attack is amplified compared to IID scenarios, particularly when Trimmed Mean is used as the defense method. As the degree of Non-IID increases, so does the amplification of the attack's effect.

We focus on Trimmed Mean, which represents vector-wise comparison methods. Trimmed Mean performs poorly in maintaining fidelity in Non-IID scenarios. The test accuracy with Trimmed Mean consistently lags behind that of FedAvg, with this difference being particularly pronounced in the CIFAR10 dataset under Label-Condition Non-IID. As the degree of data heterogeneity increases, reflected by higher  $\beta$  values, the accuracy continues to decline, indicating a negative correlation between data heterogeneity and the fidelity of Trimmed Mean.

**Tip 6: Trimmed Mean, as a defense method based on comparison to compete against attacks, is affected by the Non-IID. Compared with FedAvg, the inducement of the degree of heterogeneity from both Label-Condition Non-IID and Feature-Condition Non-IID led to worse fidelity on Trimmed Mean.**

Additionally, we focus on other defense methods, namely Norm Bound, CRFL, and FLMJR. These methods consistently outperform FedAvg in terms of accuracy, demonstrating their superior fidelity in Non-IID scenarios. While increased heterogeneity does cause accuracy declines in some cases, their performance in maintaining fidelity remains satisfactory.

**Tip 7: Norm Bound, CRFL, and FLMJR, as the defense methods which do not rely on the comparison between clients, maintain fidelity in Non-IID situations. In contrast to the standard aggregation algorithm, their accuracy loss in Non-IID scenarios is always smaller.**

The evaluated defense methods demonstrate varying levels of fidelity when applied in Non-IID scenarios. Among them, Trimmed Mean, which represents vector-wise comparison methods, performs poorly in maintaining fidelity. Compared to FedAvg, the accuracy of the global model is lower when Trimmed Mean is used. In contrast, the other defense methods perform much better, consistently maintaining accuracy comparable to that of FedAvg. Notably, CRFL and FLMJR excel, outperforming FedAvg in Feature-Condition Non-IID and Label-Condition Non-IID, respectively.

The heterogeneous data distribution among clients leads to divergence in their local models. In scenarios without attacks, this divergence can be beneficial for aggregation, as it contains information that aids the convergence and generalization of the global model. However, defense methods based on comparison may disregard the contributions of such local models. In Non-IID scenarios, the defender faces difficulty in distinguishing between model divergence caused by poisoning attacks and that arising from Non-IID, which undermines the fidelity of Trimmed Mean. In contrast, CRFL and FLMJR, which introduce perturbations to the global model and enhance the consistency of the global model to noise, respectively, are well-suited to handle model divergence induced by Non-IID, leading to their strong performance in maintaining fidelity.

### B. Impacts of Non-IID to the Robustness of Defenses

Next, we shift our focus to the robustness of defense methods in Non-IID scenarios with untargeted attacks. The defender's responsibility is to ensure the global model's functionality in FL, even in the presence of adversarial attacks.

TABLE II

TEST ACCURACY OF DEFENSE METHODS UNDER UNTARGETED ATTACKS IN IID, LABEL-CONDITION NON-IID, AND FEATURE-CONDITION NON-IID

| CIFAR10      |           |       |       |       |                         |       |       |       |                           |       |       |       |
|--------------|-----------|-------|-------|-------|-------------------------|-------|-------|-------|---------------------------|-------|-------|-------|
| Method       | IID       |       |       |       | Label-Condition Non-IID |       |       |       | Feature-Condition Non-IID |       |       |       |
|              | No attack | Fang  | NDSS  | LIE   | No attack               | Fang  | NDSS  | LIE   | No attack                 | Fang  | NDSS  | LIE   |
| FedAvg       | 72.24     | 70.39 | 71.58 | 68.13 | 68.09                   | 63.52 | 66.99 | 60.27 | 72.07                     | 68.37 | 71.06 | 64.28 |
| Trimmed Mean | 71.76     | 70.09 | 71.34 | 58.62 | 65.07                   | 63.47 | 65.22 | 46.51 | 71.53                     | 68.17 | 70.29 | 55.01 |
| Norm Bound   | 72.59     | 71.48 | 72.18 | 67.23 | 68.25                   | 67.11 | 66.41 | 60.03 | 72.04                     | 71.17 | 71.79 | 64.55 |
| CRFL         | 75.34     | 70.19 | 74.28 | 69.10 | 69.67                   | 63.53 | 68.76 | 61.18 | 74.19                     | 68.19 | 73.24 | 64.99 |
| FLMJR        | 76.59     | 74.14 | 75.64 | 72.5  | 71.36                   | 62.28 | 71.37 | 63.48 | 73.98                     | 72.22 | 73.50 | 69.35 |

| MNIST        |           |       |       |       |                         |       |       |       |                           |       |       |       |
|--------------|-----------|-------|-------|-------|-------------------------|-------|-------|-------|---------------------------|-------|-------|-------|
| Method       | IID       |       |       |       | Label-Condition Non-IID |       |       |       | Feature-Condition Non-IID |       |       |       |
|              | No attack | Fang  | NDSS  | LIE   | No attack               | Fang  | NDSS  | LIE   | No attack                 | Fang  | NDSS  | LIE   |
| FedAvg       | 98.84     | 98.53 | 98.28 | 98.37 | 98.82                   | 98.44 | 97.77 | 98.35 | 98.13                     | 97.36 | 96.68 | 97.61 |
| Trimmed Mean | 98.58     | 98.38 | 98.46 | 97.88 | 98.17                   | 97.69 | 97.74 | 96.87 | 96.91                     | 96.62 | 96.79 | 96.1  |
| Norm Bound   | 98.90     | 98.60 | 98.61 | 98.40 | 98.78                   | 98.61 | 98.64 | 98.25 | 97.96                     | 97.81 | 97.67 | 97.34 |
| CRFL         | 98.92     | 98.17 | 98.26 | 98.55 | 98.83                   | 98.14 | 97.74 | 98.27 | 98.05                     | 97.55 | 96.78 | 96.78 |
| FLMJR        | 99.06     | 98.36 | 98.74 | 99.03 | 98.92                   | 98.44 | 98.35 | 98.94 | 98.61                     | 98.27 | 97.93 | 98.17 |

As shown in the previous section, the interaction between untargeted attacks and Non-IID exacerbates the damage to FL, making it challenging to maintain the robustness of defense methods. Therefore, this section investigates the robustness of defense methods against untargeted attacks in Non-IID scenarios. We directly estimate the test accuracy of the global model in FL across different conditions to assess the performance of these defense methods.

Similar to the last experiment, for this evaluation, we select the CIFAR10 dataset and MNIST dataset and estimate the performance of Trimmed Mean, Norm Bound, CRFL, FLMJR, and FedAvg. We evaluate the performance of each method in three scenarios: IID, Label-Condition Non-IID, and Feature-Condition Non-IID. In the CIFAR10 dataset and MNIST dataset, we set  $\beta = 0.2$  and  $\beta = 0.4$  to simulate Non-IID respectively. LIE, Fang, and NDSS are utilized as the untargeted attacks. 10% of the total clients are set as adversaries. The results are shown in Table II.

Regarding the performance of Trimmed Mean, it is clear that this method does not offer satisfactory robustness. The introduction of data heterogeneity consistently leads to a decline in accuracy across all scenarios. Particularly in the CIFAR10 dataset, its performance against LIE is the poorest, with a significantly greater accuracy drop compared to other attack scenarios. In the MNIST dataset, Trimmed Mean still underperforms relative to other defense methods as well as the standard aggregation method.

**Tip 8: Similar to the results of fidelity, Trimmed Mean demonstrates poor robustness in Non-IID scenarios. Across both Label-Condition Non-IID and Feature-Condition Non-IID, its accuracy falls below that of FedAvg, indicating a compromised global model functionality under untargeted attacks.**

Furthermore, we discuss the performance of the remaining defense methods. These methods have demonstrated robustness in Non-IID scenarios. Although the introduction of data heterogeneity still leads to a decline in accuracy, the

resulting accuracy remains higher than that of FedAvg in most cases, indicating that these methods maintain robustness in both Feature-Condition Non-IID and Label-Condition Non-IID scenarios. Notably, FLMJR achieves the highest accuracy in most scenarios, underscoring its strong robustness against untargeted attacks in Non-IID settings.

**Tip 9: Norm Bound, CRFL, FLMJR provide satisfactory in maintaining the accuracy under untargeted attacks in both Feature-Condition Non-IID and Label-Condition Non-IID which demonstrate their robustness in Non-IID scenarios.**

The defense methods exhibit varying levels of robustness in Non-IID scenarios. As with the fidelity evaluation, Trimmed Mean performs poorly against untargeted attacks, with test accuracy lower than that of FedAvg. In contrast, Norm Bound and CRFL demonstrate satisfactory performance, achieving better test accuracy than FedAvg in most cases. FLMJR stands out by achieving the highest robustness against untargeted attacks, consistently yielding the highest test accuracy in both Label-Condition Non-IID and Feature-Condition Non-IID scenarios.

The poor robustness of Trimmed Mean in Non-IID scenarios arises from its reliance on direct comparisons among local models to identify and eliminate malicious models. This method assumes that malicious models will exhibit significant divergence from benign models. However, Non-IID amplifies the divergence among local models from benign clients, confusing the defender and hindering the detection of malicious models, thereby contributing to Trimmed Mean's poor performance. In contrast, CRFL and Norm Bound defend against poisoning attacks by bounding L2 norms (CRFL bounds the global model's norm, while Norm Bound bounds the norms of local models). Although Non-IID induces divergence among local models, making it difficult to establish a reliable bounding standard, these defense methods still perform satisfactorily when an appropriate standard is set. FLMJR, which strengthens FL robustness by injecting model Jacobian regularization

into clients' local objectives, is minimally affected by Non-IID. Moreover, the divergence among local models induced by Non-IID can be viewed as a negative perturbation to the aggregated global model. FLMJR mitigates this issue by enhancing the stability of the global model against such perturbations.

## VII. DISCUSSION

Building on our empirical findings regarding the relationship between data heterogeneity and security in FL, we further analyze the results to propose an optimal solution for defending against untargeted attacks.

### A. Untargeted Attack

The objective of untargeted attacks in FL is to bypass defense methods and induce utility loss in the global model. This objective can be broken down into two key attributes that adversaries aim to achieve: stealthiness and harmfulness. However, as the results show, the evaluated attacks do not fully align with these attributes in Non-IID scenarios. A key reason for this is that these attacks are designed to disrupt the convergence of the global model, which becomes increasingly difficult in the presence of benign clients. Moreover, the manipulation of the malicious model can be easily detected by monitoring the training loss. As a natural challenge, Non-IID increases the difficulty of convergence and leads to a decline in accuracy. We identify a potential research direction for untargeted attacks: to disguise the attack as Non-IID. In this approach, the attack can exploit Non-IID to maintain stealthiness, while simultaneously causing significant damage to FL, much like Non-IID itself.

### B. Defense Method

In Non-IID scenarios, defense methods have two main objectives: fidelity and robustness. Fidelity refers to maintaining the functionality of the global model in the absence of attacks, while robustness aims to mitigate the effects of poisoning attacks. Our results demonstrate that defense methods relying on local model comparison struggle in Non-IID scenarios. The heterogeneity introduced by Non-IID in the local model distribution complicates the defenders' ability to distinguish between malicious models. Despite their subpar performance in our evaluation, we believe comparison-based methods remain a valid approach for defense, as they aim to reduce the impact of adversaries. For future work on comparison-based defense methods, we suggest focusing on differentiating the effects of malicious models from those caused by Non-IID, which is particularly crucial in practice due to the widespread occurrence of Non-IID in FL production environments, posing a significant threat to the utility of defense methods.

## VIII. CONCLUSION

In this study, we use conditional distribution to characterize data heterogeneity in FL, defining two primary types of Non-IID: Label-Condition Non-IID and Feature-Condition

Non-IID. We introduce a data partitioning algorithm to simulate these Non-IID scenarios, enabling quantifiable analysis. Through extensive experiments in a cross-device environment, we empirically assess the relationship between data heterogeneity and security in FL. Our evaluation results offer valuable insights into the strengths and weaknesses of existing methods from both attacker and defender perspectives. We hope our study will inspire the research community to tackle the practical security challenges in FL.

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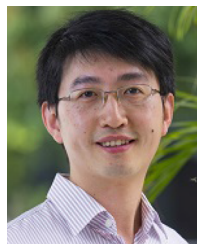
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